Technology Stack

Date	04 july 2026
Team ID	xxxxxx
Project Name	Human Development Index(HDI) Predictor
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	**Architecture Component / Layer**	**Technology Chosen**	**Justification / Purpose**
1	**Frontend / Client-Side** *(e.g., React, Angular, HTML/CSS)*	*HTML, CSS, JavaScript*	*Provides a responsive and user-friendly interface for entering HDI data, viewing predictions, and interacting with the portal.*

2	**Backend / Server-Side** *(e.g., Node.js, Python/Django, Java)*	*Python (Flask)*	*Flask handles user requests, processes input data, integrates the machine learning model, and generates HDI predictions.*
3	**Database / Data Storage** *(e.g., MySQL, MongoDB, PostgreSQL)*	*CSV Dataset / Pandas Data Frames*	*Stores and processes HDI data efficiently for training and prediction without requiring a full database.*
4	**Cloud / Hosting / Deployment** *(e.g., AWS, Azure, Heroku, Docker)*	*Render*	*Hosts and deploys the HDI Prediction web application online, making it accessible through a web browser.*
5	**Version Control & CI/CD** *(e.g., GitHub, GitLab, Jenkins)*	GitHub	*Maintains source code, tracks changes, and supports collaboration & deployment.*
6	**Third-Party APIs / Other Tools** *(e.g., Stripe, Twilio, Google Maps)*	Scikit-learn, Pandas, NumPy, Matplotlib	*Used for machine learning model development, data preprocessing, prediction, and visualization.*